

30 2026-04-20 | Week 14 | Lecture 30

30.1 Decision errors

Example 134 (Coin flipping hypothesis test). Flip a coin 10 times. Let p be the probability of heads. Consider the hypothesis testing problem

$$H_0 : p = \frac{1}{2} \quad \text{versus} \quad H_a : p \neq \frac{1}{2}.$$

Let our test statistic be

$$T = (\# \text{ of heads})$$

So T takes a value between 0 and 10. If H_0 were true, then (one can check that) there would be about 97.9% chance of getting between 2 and 8 heads. So it might be reasonable to define our rejection region to be the set

$$\text{RR} = \{0, 1, 9, 10\}$$

since it is improbable to get these values with a fair coin. Using this rejection region, if $T = 0, 1, 9$ or 10 , we would reject H_0 and conclude the coin is not a fair coin.

This test is not perfect: assuming that H_0 is true, we will nonetheless reject H_0 about 2.1% of the time.

End of Example 134. $\square$

30.1.1 Type I and Type II errors

There are two types of decision errors in the hypothesis testing framework.

	Reject H_0	Accept H_0
H_0 true	Type I error level = $\mathbb{P}[\text{type I error}] = \alpha$	Correctly accept H_0 $1 - \alpha$
H_a true	Correctly reject H_0 power = $1 - \beta$	Type II error $\mathbb{P}[\text{type II error}] = \beta$

Example 135 (Example 134 continued). In Example 134, the level of the test is $\alpha = 2.1\%$. Note that β is a function of p . For example if $p = 0.4$, then

$$\begin{aligned} \beta &= \beta(0.4) \\ &= \mathbb{P}[\text{Type II error} \mid p = 0.4] \\ &= \mathbb{P}[T \notin \text{RR} \mid p = 0.4] \\ &= \sum_{k=2}^8 \binom{10}{k} (0.4)^k (1 - 0.4)^{10-k} \\ &= .952 \end{aligned}$$

Therefore if the true value of p is .4, then the power of the the test is .048.

On the other hand if the true value of p is $p = .1$, then $\beta(.1) = .264$, so the power of the test would be .736. We generally don't know what the true power of the test is, since that would require knowing the value of the parameter (in which case we wouldn't be doing a hypothesis test...).

It is a common mistake to assume that $\alpha + \beta = 1$. In general, $\alpha + \beta \neq 1$.

End of Example 135. $\square$

These terms are clarified by an example.

Remark 136 (Decision Errors). A Ukrainian drone operator spots an unmarked vehicle traveling down a road. She needs to make a decision about whether to strike the truck. The hypothesis test is the following:

$$\begin{aligned} H_0 &: \text{the target is not a Russian military vehicle} \\ H_a &: \text{the target is a Russian military vehicle} \end{aligned}$$

Rejecting H_0 means giving an order to destroy the target. Accepting H_0 means ‘take no action’.

In this context, a Type-I error occurs when the operator gives the order to destroy the vehicle, but it turns out the vehicle is not Russian military. A Type-II error, on the other hand, has occurred if she doesn’t order the strike, but the vehicle really was a Russian military vehicle.

- A Type-I error means she killed a civilian or a friendly unit (an error of commission).
- A Type-II error means she missed an opportunity to kill an enemy (an error of omission).

Here, the terms “power” and “level” have the following meaning:

- The **power** of the test is the probability that she orders a strike when a Russian military vehicle goes by. (She wants this to be high).
- The **level** of the test is the probability that she orders a strike when a civilian goes by. (She wants this to be low).

Ideally, the operator would like to eliminate both types of errors (accidentally killing civilians vs failing to kill enemy). That is, she would like

$$\alpha = \Pr[\text{Type I error}] = 0 \quad \text{and} \quad \beta = \Pr[\text{Type II error}] = 0.$$

Unfortunately, it is impossible to simultaneously eliminate both types of error without perfect information. On one hand, the drone operator could minimize the probability of a type I error by simply never ordering a strike, in which case $\alpha = \Pr[\text{Type I error}] = 0$. But this is unsatisfactory, since in that case, $\beta = \Pr[\text{Type II error}] = 1$. On the other hand, she could maximize the power by simply always ordering a strike; in that case, $\beta = 0$ but $\alpha = 1$. This tradeoff is present in every hypothesis test.

The approach that the hypothesis testing framework takes to address this tradeoff is to (1) set a maximal level α that one is willing to accept, and then (2) find the most powerful test subject to that constraint.

Of course, the appropriate significance level to set depends on the situation: a truck speeding towards the operator’s position is very different from a truck traveling away from the front.